

# A self-supervised electrocardiogram foundation model for empowering cardiovascular disease prediction and genetic factor discovery

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Electrocardiogram (ECG) has been widely used in the diagnosis of cardiovascular disease (CVD). Current deep learning methods for CVD prediction using ECG often lack generalizability and interpretability, resulting in limited performance. Here, we have developed a self-supervised Electrocardiogram Large-scale Foundation Model (ECG-LFM) through pre-training over ten million 12-lead ECGs from multiple ECG datasets. To enhance ECG representation, ECG-LFM integrates contrastive learning with masked language modeling in a self-supervised manner, enabling the model to capture both global contextual information and fine-grained patterns within ECG signals. It was fine-tuned to predict eight types of CVDs and achieved an average area under the receiver operating characteristic curve (AUROC) of 0.930 from multiple datasets, which demonstrates improved performance compared to existing methods. The important ECG-LFM derived features (EDFs) are able to represent known CVD biomarkers, indicating the high interpretability of ECG-LFM. Applications of the EDFs in genome-wide association study identified 24 significant single nucleotide polymorphisms (SNPs) ( $P$ -value  $< 5 \times 10^{-8}$ , LD  $r^2 < 0.01$ ) associated with ECG, including 8 novel findings. The genetic causal effects of EDFs on the CVDs were evaluated by Mendelian randomization, indicating 2 CVDs and 4 EDFs having causal relationships. Overall, ECG-LFM provides accurate prediction for CVDs and novel genetic insights for ECG.

Cardiovascular disease (CVD) is a leading cause of death worldwide, necessitating early and accurate diagnosis<sup>1</sup>. The most frequently used tool for CVD assessment is the electrocardiogram (ECG), a non-invasive and cost-effective technique that has been widely employed in clinical practice<sup>2</sup>. However, traditional methods for ECG analysis, such as the Glasgow algorithm<sup>3</sup>, often lack the flexibility to capture complex

spatiotemporal relationships in ECG signals, resulting in limited CVD risk prediction.

Recently, deep learning techniques have been used in ECG analysis and have achieved superior performance. For example, A. Sau et al. employed a convolutional neural network to predict mortality and cardiovascular risk through ECG<sup>4</sup>. T. Tohyama et al. infer 5-year

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incidence of atrial fibrillation (AF) using 12-lead ECGs through a deep learning technique<sup>5</sup>. However, training these models requires manually labeled high-quality data, which poses significant challenges in terms of scalability, as obtaining such data is time-consuming. More importantly, these models lack generalizability for various diagnostic tasks and fail to fully leverage the rich information embedded in large ECG datasets, highlighting the need for novel computational strategies.

Foundation models have recently shown remarkable generalization abilities in various medical tasks by leveraging large-scale pre-training and transfer learning paradigms. For example, the Clinical Histopathology Imaging Evaluation Foundation model provides a generalizable foundation to efficiently evaluate digital pathology for cancer patients<sup>6</sup>. In retinal disease diagnosis, the RETFound model has achieved excellent performance across various clinical diagnostic tasks through pre-training on a large number of retinal images<sup>7</sup>. In the field of ECG analysis, foundation models are also beginning to demonstrate their potential. For example, HeartBEiT<sup>8</sup> leverages a vision-based transformer to capture both the temporal and spatial features of ECG signals. Besides, KED utilizes GPT-4 to incorporate domain-specific knowledge of ECG signals by training on approximately 800 thousand ECGs<sup>9</sup>. However, these ECG foundation models still face common limitations. Firstly, while some recent foundation models like HeartBEiT have been trained on substantial datasets (over 8 million ECGs), the potential of models pre-trained on more than 10 million ECGs remains to be fully explored. Secondly, the predictions of these models lack interpretability, resulting in a fundamental barrier to adoption by cardiologists who demand interpretable decision support for cardiac diagnoses. Besides, they fail to capture the relationships between genotype and ECG representations from the pre-training, limiting their ability to identify genetic variations associated with ECG and CVD. Therefore, it is critical to develop an interpretable ECG foundation model that leverages larger datasets, provides accurate predictions, and uncovers genetic insights for various CVDs.

In this study, we have proposed the Electrocardiogram Large-scale Foundation Model (ECG-LFM), a general-purpose self-supervised framework that provides the foundation for various CVD diagnosis and prediction tasks. We performed self-supervised pre-training for ECG-LFM using over ten million 12-lead ECGs from two large-scale ECG datasets, the Harvard-Emory ECG Database (HEEDB) and the MIMIC-IV-ECG database. ECG-LFM extracts the latent representations (tokens) from raw 12-lead ECG inputs by a feature encoder, discretizes the tokens to obtain the target context vectors and discretized IDs by a quantization module, and extracts contextual representations for each latent representation by an embedding module. Finally, ECG-LFM employs a multi-task learning strategy, combining contextual contrastive loss, masked language modeling (MLM) loss, and multi-segment contrastive loss to enhance representation learning. Our results demonstrate that ECG-LFM significantly outperforms state-of-the-art methods in predicting various CVDs in multiple datasets (PTB-XL, Chapman, CODE-15, CPSC-2018, and UK Biobank), and has achieved an average area under the receiver operating characteristic curve (AUROC) of 0.930. The performance of ECG-LFM has not been influenced, even if it is fine-tuned by a set of samples with a much smaller size. To enhance model interpretability, we further evaluated the contributions of ECG-LFM-derived features (EDFs) in CVD prediction and detected the biological connections of EDFs with ECG morphologies through the perturbation-based EDF interpretation (PEI) method. The phenotypic relationships of the EDFs showing importance in predicting CVDs with 215 cardiac function-related factors were examined by the Phenotype-wide association study (PheWAS), indicating that these EDFs are significantly associated with 167 factors, especially the physical measures and the features of cardiac MRI. The genetic characteristics of the important EDFs were evaluated by genome-wide association studies (GWAS) that have identified 12 single-nucleotide polymorphisms (SNPs) ( $P$  value  $< 5 \times 10^{-8}$ , LD

$r^2 < 0.01$ ) at 11 loci. The genetic causal relationships of these EDFs with the CVDs were evaluated by Mendelian randomization (MR) analyses, indicating that 4 EDFs and 2 CVDs had causal relationships. GWAS for all 1024 EDFs identified 8 new significant SNPs associated with ECG. Thus, ECG-LFM is a highly generalizable self-supervised learning foundation model for ECG analysis, which has shown significant improvement in predicting CVDs. Features derived by ECG-LFM have improved our understanding of phenotypic and genetic characteristics of ECG in cardiac abnormalities.

## Results

### Overview of this study

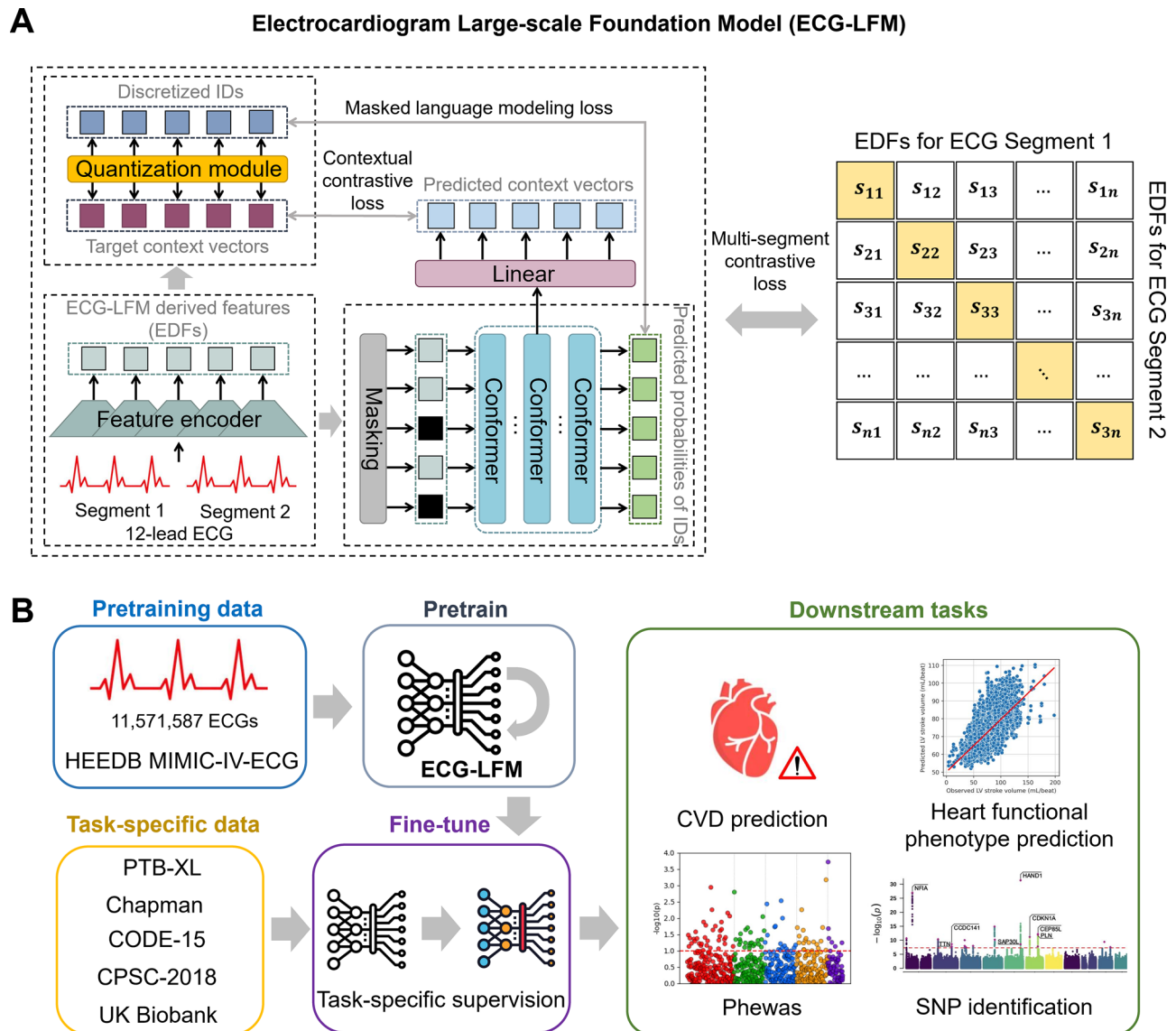
Figure 1 depicts the overall workflow of this study. We collected ECGs from the HEEDB<sup>10</sup> and MIMIC-IV-ECG dataset<sup>11</sup>, leading to a total of 11,571,587 12-lead ECG records of 1,979,599 individuals (Supplementary Data 1 and Supplementary Fig. S1) for model pre-training. By using this large ECG dataset, we constructed ECG-LFM, an ECG foundation model that captures intricate patterns of signals, providing the foundation for various CVD prediction tasks. ECG-LFM converts 12-lead ECG data into a series of latent representations (tokens) through a feature encoder. These tokens are discretized into context vectors and indexes (IDs) by the quantization module for efficient encoding. ECG-LFM then masks the tokens from the feature encoder and extracts contextual information from ECGs through multiple conformer blocks. The context vectors and the discretized IDs of the masked tokens were predicted by ECG-LFM using the contextual information from the surrounding tokens through solving a contextual contrastive task and an MLM task. To further improve ECG-LFM, two segments of ECG were taken as input, and the multi-segment contrastive loss was used to learn consistent representations of consecutive information in each ECG (Fig. 1A).

Once pre-trained, ECG-LFM was then fine-tuned for CVD risk and heart functional phenotype predictions (Fig. 1B). The genetic and phenotypic interpretations of the ECG-LFM-derived features (EDFs) showing important contributions in predicting risks of CVDs were investigated by GWAS and PheWAS. The ability of EDFs in empowering the identification of genetic variants associated with ECG was examined by GWAS for the top 10 components of all EDFs.

### Pre-training of ECG-LFM provides individual-specific and disease-relevant ECG representations

To investigate whether the pre-training of ECG-LFM provides ECG representations that capture individual-specific biometric patterns, we evaluated the performance of the pre-trained model on individual identification through linear probing. Briefly, from the feature encoder of the pre-trained ECG-LFM, 1024-dimensional embeddings were extracted, which were used to characterize the individuals through training a single-layer perceptron. The extracted embeddings have achieved an average Top-3 accuracy (Top-3 ACC) of 0.65 in recognizing the individuals in the Chapman and CPSC-2018 datasets (Fig. 2A). ECG-LFM is compared to HeartBEiT<sup>8</sup> that is the only publicly available model generating the embeddings. The embeddings generated by HeartBEiT have achieved Top-3 ACC = 0.43, which is a significant (one-sided  $t$ -test  $P$  value =  $3.81 \times 10^{-22}$ ) decrease compared to ECG-LFM. To visualize the individual identification performed by ECG-LFM and HeartBEiT, we randomly selected 100 individuals aged from 40 to 60 from the CPSC-2018 dataset and characterized them by embeddings through  $t$ -distributed stochastic neighbor embedding ( $t$ -SNE) (Fig. 2B). The embeddings generated by ECG-LFM are more well-separated with adjusted rand index (ARI) = 0.73 than HeartBEiT (ARI = 0.41). This result demonstrates that pre-training of ECG-LFM captures more individual-specific information from ECG than HeartBEiT.

To further evaluate the embeddings extracted from pre-trained ECG-LFM and HeartBEiT in disease relevance, we performed the cardiac arrhythmia prediction for individuals in the Chapman and CPSC-



**Fig. 1 | Overview of the study. A** We constructed the Electrocardiogram Large-scale Foundation Model (ECG-LFM), a general-purpose machine learning framework that provides the foundation for various CVD diagnosis and prediction tasks. ECG-LFM is composed of three modules. The first one is a feature encoder comprising multiple convolution blocks, which converts pre-processed 12-lead ECG inputs into a series of latent representations (tokens). These tokens are discretized into context vectors and indexes (IDs) by the second quantization module. In the third module, ECG-LFM randomly masks a portion of the tokens from the feature encoder and extracts contextual information from the tokens through multiple conformer blocks. ECG-LFM predicts the context vectors by solving a contextual contrastive loss, and predicts the discretized IDs of the masked tokens by solving a masked language modeling (MLM) loss. ECG-LFM is further improved by using a multi-segment contrastive loss model to learn consistent representations of

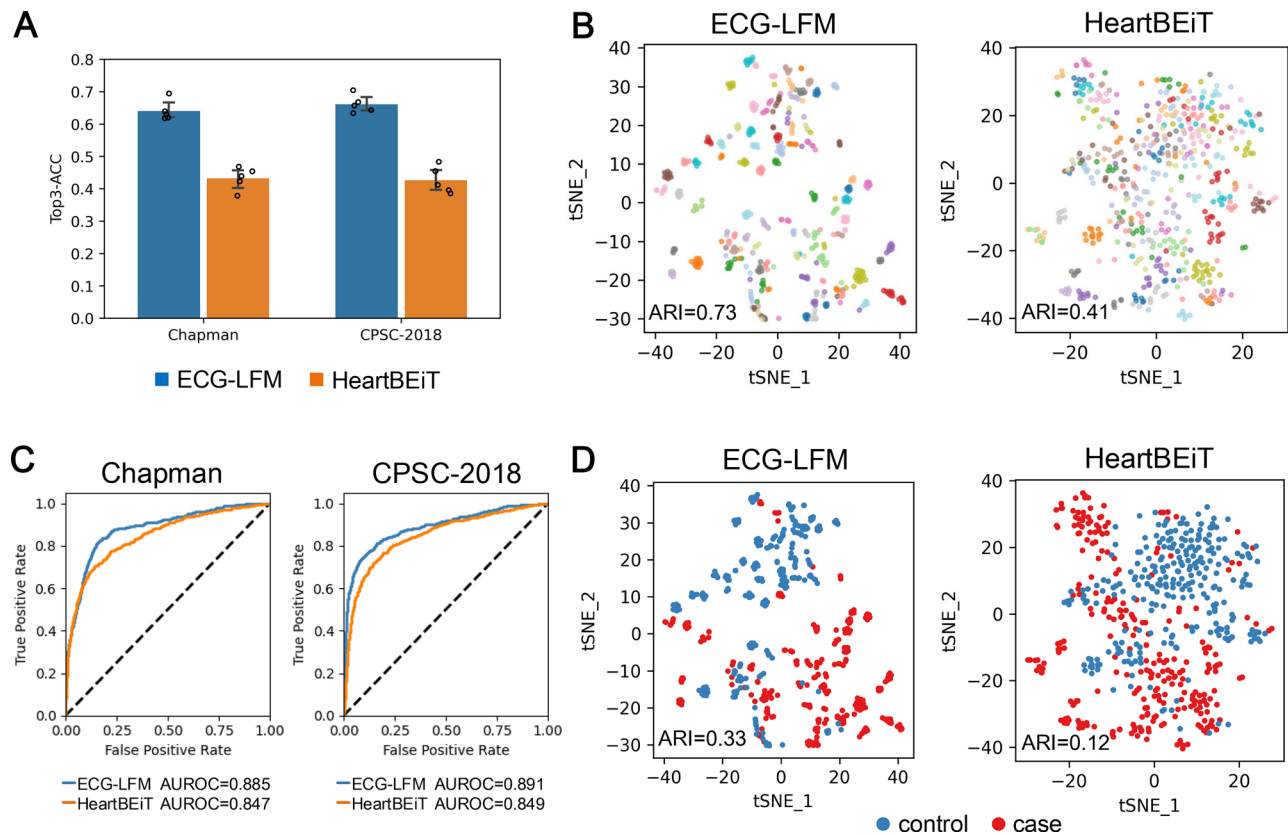
consecutive information from ECG segments. **B** We collected a total of 11,571,587 12-lead ECG records from Harvard-Emory ECG Database (HEEDB) and MIMIC-IV-ECG database for model pre-training. The performance of ECG-LFM was evaluated for a wide range of ECG evaluation tasks in five task-specific ECG datasets (PTB-XL, Chapman, CODE-15, CPSC-2018, and UKB). The prediction tasks are cardiovascular disease (CVD) prediction and heart functional phenotype prediction. The importance of the ECG-LFM-derived features (EDFs) was evaluated by the integrated gradient (IG) and perturbation-based EDF interpretation (PEI) method. The phenotypic and genetic correlations between EDFs and heart functions were evaluated by phenotype-wide association study (PheWAS) and genome-wide association study (GWAS). Icons used in this figure were sourced from Icons8 (<https://icons8.com>).

2018 datasets. The embeddings of ECG-LFM achieved AUROC of 0.885 and 0.891 in two independent test sets derived from Chapman and CPSC-2018, respectively (Fig. 2C). In comparison, HeartBEiT achieved an AUROC of 0.847 and 0.849 in these two independent test sets (Fig. 2C). The superior performance of ECG-LFM was further substantiated by visualizing embeddings of cases and controls using t-SNE (ARI = 0.33, Fig. 2D). In comparison, HeartBEiT exhibited an overlap in the embedding space that could undermine cardiac arrhythmia prediction accuracy (ARI = 0.12). In summary, these results highlight the

superior capability of ECG-LFM in capturing both individual-specific biometric patterns and disease-relevant ECG representations.

### Fine-tuning ECG-LFM for predicting cardiovascular diseases and heart functions

To evaluate the performance of ECG-LFM in predicting risks of CVDs, we fine-tuned the model for predicting eight types of CVDs that are conduction disturbance (CD), hypertrophy (HYP), myocardial infarction (MI), ST/T change (STTC), ventricular ectopics (VE), sinus



**Fig. 2 | Comparing the performance of the pre-trained models through linear probing.** **A** Comparing the performance of ECG-LFM and HeartBEiT on the individual identification task. The performance is evaluated by Top-3 accuracy (Top-3 ACC). Data are presented as mean values  $\pm$  SD ( $n = 5$  replicates). **B** The visualization results of ECG-LFM and HeartBEiT are plotted by the t-distributed stochastic neighbor embedding (t-SNE) using the ECG embeddings from 100 randomly selected individuals aged from 40 to 60 of the CPSC-2018 dataset. Each point represents the embedding of an ECG segment, and the color of each point corresponds to a unique individual. The adjusted rand indexes (ARI) for ECG-LFM and

HeartBEiT are 0.73 and 0.41, respectively. **C** Comparing the performance of ECG-LFM and HeartBEiT on the cardiac arrhythmia prediction task. The performance is evaluated by the area under the receiver operating characteristic curve (AUROC). **D** The visualization embeddings of ECG-LFM and HeartBEiT are plotted by t-SNE using the ECG embeddings from 100 randomly selected individuals aged from 40 to 60 of the CPSC-2018 dataset. The blue spots represent the health controls, and the red spots represent the cases. The ARI values for ECG-LFM and HeartBEiT are 0.33 and 0.12, respectively. Each point represents the embedding of an ECG signal. The ECG signals in blue are from healthy controls, and in red are from the patient.

bradycardia (SB), atrial fibrillation (AFIB), and general supraventricular tachycardia (GSVT) for individuals from the PTB-XL and Chapman datasets. Additionally, to evaluate its generalizability across multiple datasets, ECG-LFM was fine-tuned by the ECG data from the PTB-XL and Chapman datasets, and was independently tested by the ECG data from the CODE-15, CPSC-2018, and UK Biobank (UKB) datasets. The number of samples in each dataset is shown in Supplementary Data 1.

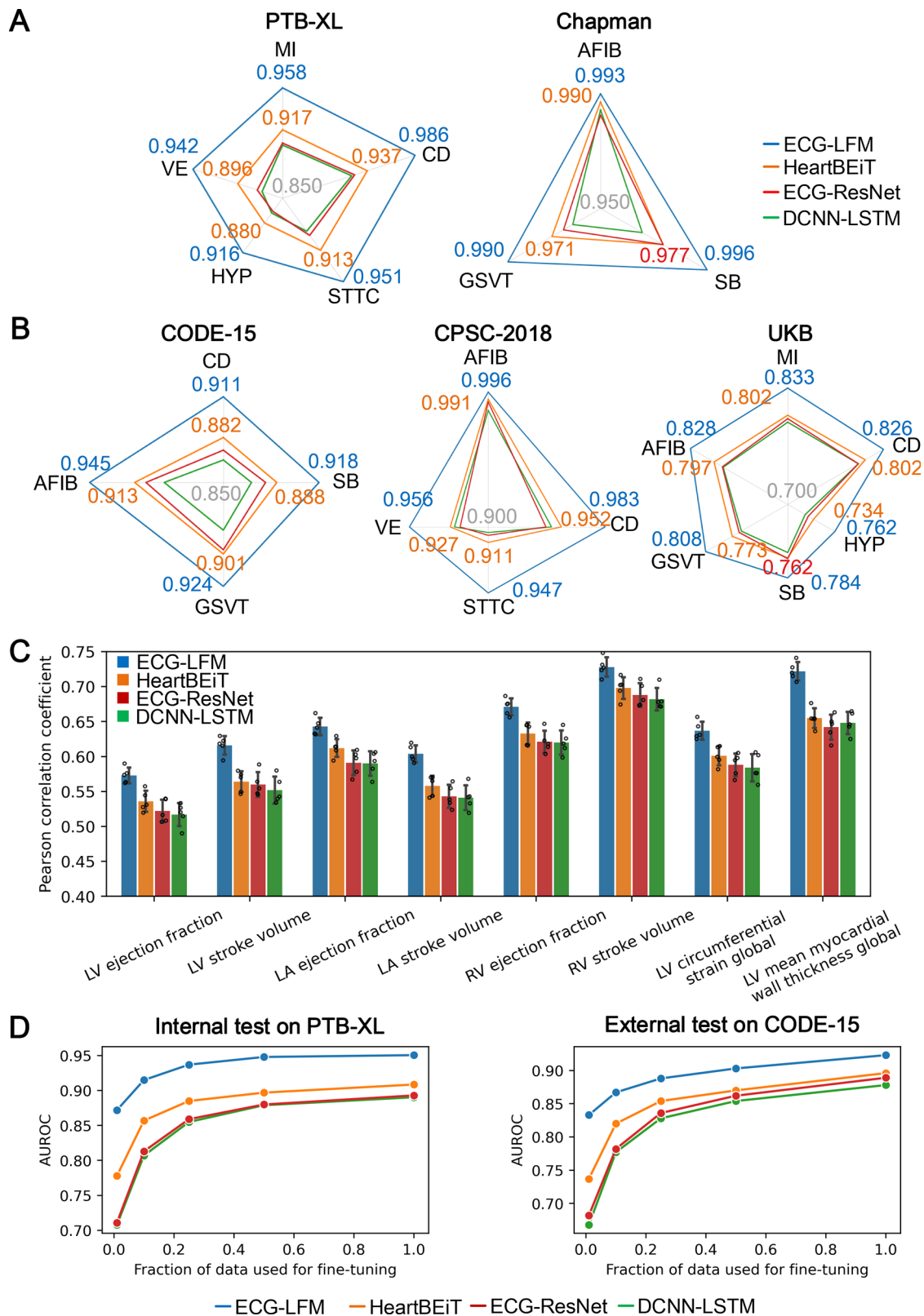
On the PTB-XL dataset, ECG-LFM achieved average AUROC of 0.951 in the independent test (ranging from 0.916 to 0.986) for predicting conduction disturbance (CD) (AUROC = 0.986), hypertrophy (HYP) (AUROC = 0.916), myocardial infarction (MI) (AUROC = 0.958), ST/T change (STTC) (AUROC = 0.951) and ventricular ectopics (VE) (AUROC = 0.942) when it was fine-tuned by the ECGs of 14,495 individuals (80%) and was tested by the ECGs of 3623 (20%) individuals (Fig. 3A and Supplementary Fig. S2). This performance significantly (two-sided  $t$ -test  $P$  value  $< 4.55 \times 10^{-24}$ ) surpasses other state-of-the-art methods, HeartBEiT<sup>8</sup> (average AUROC = 0.908), ECG-ResNet<sup>12</sup> (average AUROC = 0.893), and DCNN-LSTM<sup>13</sup> (average AUROC = 0.890) with an increase of 4.6, 6.5, and 6.8%, respectively.

When ECG-LFM was fine-tuned by ECGs of 51,297 individuals (80%) and was tested by 12,824 individuals (20%) from Chapman dataset, it achieved an average AUROC of 0.993 in the independent test set (ranging from 0.990 to 0.996) for predicting sinus bradycardia (SB) (AUROC = 0.996), atrial fibrillation (AFIB) (AUROC = 0.993) and general supraventricular tachycardia (GSVT) (AUROC = 0.990). The

performance is significantly better than other methods (two-sided  $t$ -test  $P$  value  $< 8.88 \times 10^{-6}$ ) (Fig. 3A and Supplementary Fig. S2).

The ECG-LFM was further fine-tuned by using PTB-XL and Chapman datasets as the training set, and independently tested by the external test sets, CODE-15, CPSC-2018, and UKB, respectively. ECG-LFM achieved an average AUROC of 0.925 for predicting four types of CVDs in CODE-15, an average AUROC of 0.971 for predicting four types of CVDs in CPSC-2018, and an average AUROC of 0.807 for predicting six types of CVDs in UKB (Fig. 3B). The performance is significantly better than the existing methods, HeartBEiT (average AUROC<sub>CODE-15</sub> = 0.896, average AUROC<sub>CPSC-2018</sub> = 0.945, and average AUROC<sub>UKB</sub> = 0.778), ECG-ResNet (average AUROC<sub>CODE-15</sub> = 0.889, average AUROC<sub>CPSC-2018</sub> = 0.939, and average AUROC<sub>UKB</sub> = 0.772), and DCNN-LSTM (average AUROC<sub>CODE-15</sub> = 0.878, average AUROC<sub>CPSC-2018</sub> = 0.939, and average AUROC<sub>UKB</sub> = 0.768) (Fig. 3B). Table 1 shows the AUROC of all these models in predicting each type of CVD, indicating that ECG-LFM shows better performance in predicting all these CVDs than others.

Finally, ECG-LFM was further fine-tuned to predict eight heart functional phenotypes, including left ventricular (LV) ejection fraction, LV stroke volume, left atrial (LA) ejection fraction, LA stroke volume, right ventricular (RV) ejection fraction, RV stroke volume, LV circumferential strain global and LV mean myocardial wall thickness global by using 32,988 individuals as training set (80%) from UKB. It was independently tested by using 8247 individuals (20%) from the UKB dataset. ECG-LFM achieved an average Pearson correlation coefficient



(PCC) of 0.649, which is significantly ( $P$  value  $< 7.74 \times 10^{-4}$ ) better than HeartBEiT (average PCC = 0.607), ECG-ResNet (average PCC = 0.594), and DCNN-LSTM (PCC = 0.592) (Fig. 3C, Supplementary Fig. S4).

#### Influence of fine-tuning dataset on ECG-LFM performance

To investigate the influence of fine-tuning the dataset on ECG-LFM, the ECG-LFM was evaluated in predicting eight types of CVDs after it was

fine-tuned by the training data with varied sizes. The sizes of the training data ranged from 1%, 10%, 25%, 50%, to 100% of the PTB-XL training set, respectively. When ECG-LFM was fine-tuned by 1% of the training set and evaluated by the CVDs of 3623 individuals from the PTB-XL test set, the model achieved an average AUROC of 0.872, outperforming HeartBEiT (average AUROC of 0.778) by 12.1%, ECG-ResNet (average AUROC of 0.711) by 22.6% and DCNN-LSTM (average

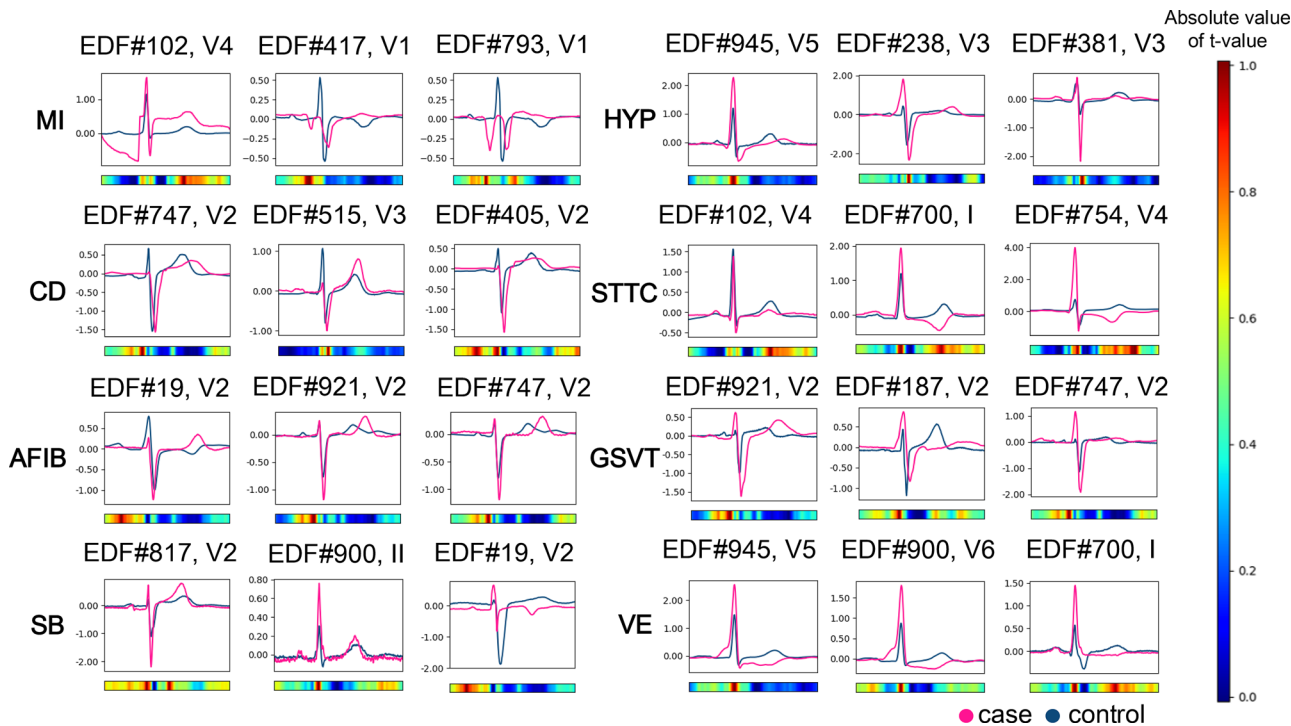
**Fig. 3 | Predicting CVDs and heart functional phenotypes by ECG-LFM, HeartBEiT, ECG-ResNet, and DCNN-LSTM.** Comparing the performance of ECG-LFM, HeartBEiT, ECG-ResNet, and DCNN-LSTM in predicting CVDs for individuals from **A** PTB-XL, Chapman, **B** CODE-15, CPSC-2018, and UKB datasets. The performance is evaluated by the area under the receiver operating characteristic curve (AUROC). The AUROC values achieved by the best two methods are shown in the panel. MI myocardial infarction, HYP hypertrophy, CD conduction disturbance, STTC ST/T change, AFIB atrial fibrillation, GSVT general supraventricular tachycardia, SB sinus bradycardia, VE ventricular ectopics. **C** Comparing the average pearson correlation

coefficient (PCC) achieved by ECG-LFM to other methods in predicting left ventricular (LV) ejection fraction, LV stroke volume, left atrial (LA) ejection fraction, LA stroke volume, right ventricular (RV) ejection fraction, RV stroke volume, LV circumferential strain global and LV mean myocardial wall thickness global in the UKB test set. The PCC is the correlation between the observed and the predicted traits. Data are presented as mean values  $\pm$  SD ( $n = 5$  replicates). **D** Comparing the AUROC performance of ECG-LFM, HeartBEiT, ECG-ResNet, and DCNN-LSTM in predicting CVDs by training the models using 1%, 10%, 25%, 50%, and 100% of samples in the training set.

**Table 1 | Performance of ECG-LFM, HeartBEiT, ECG-ResNet, and DCNN-LSTM in predicting risks of CVDs**

| Model      | PTB-XL | Chapman | CODE-15 | CPSC-2018 | UKB   | AVG   |
|------------|--------|---------|---------|-----------|-------|-------|
| ECG-LFM    | 0.951  | 0.993   | 0.925   | 0.971     | 0.807 | 0.930 |
| HeartBEiT  | 0.909  | 0.979   | 0.896   | 0.945     | 0.778 | 0.901 |
| ECG-ResNet | 0.893  | 0.976   | 0.889   | 0.939     | 0.772 | 0.894 |
| DCNN-LSTM  | 0.890  | 0.972   | 0.878   | 0.939     | 0.768 | 0.889 |

The performance is evaluated by the area under the receiver operating characteristic curve (AUROC).  
AVG average AUROC.

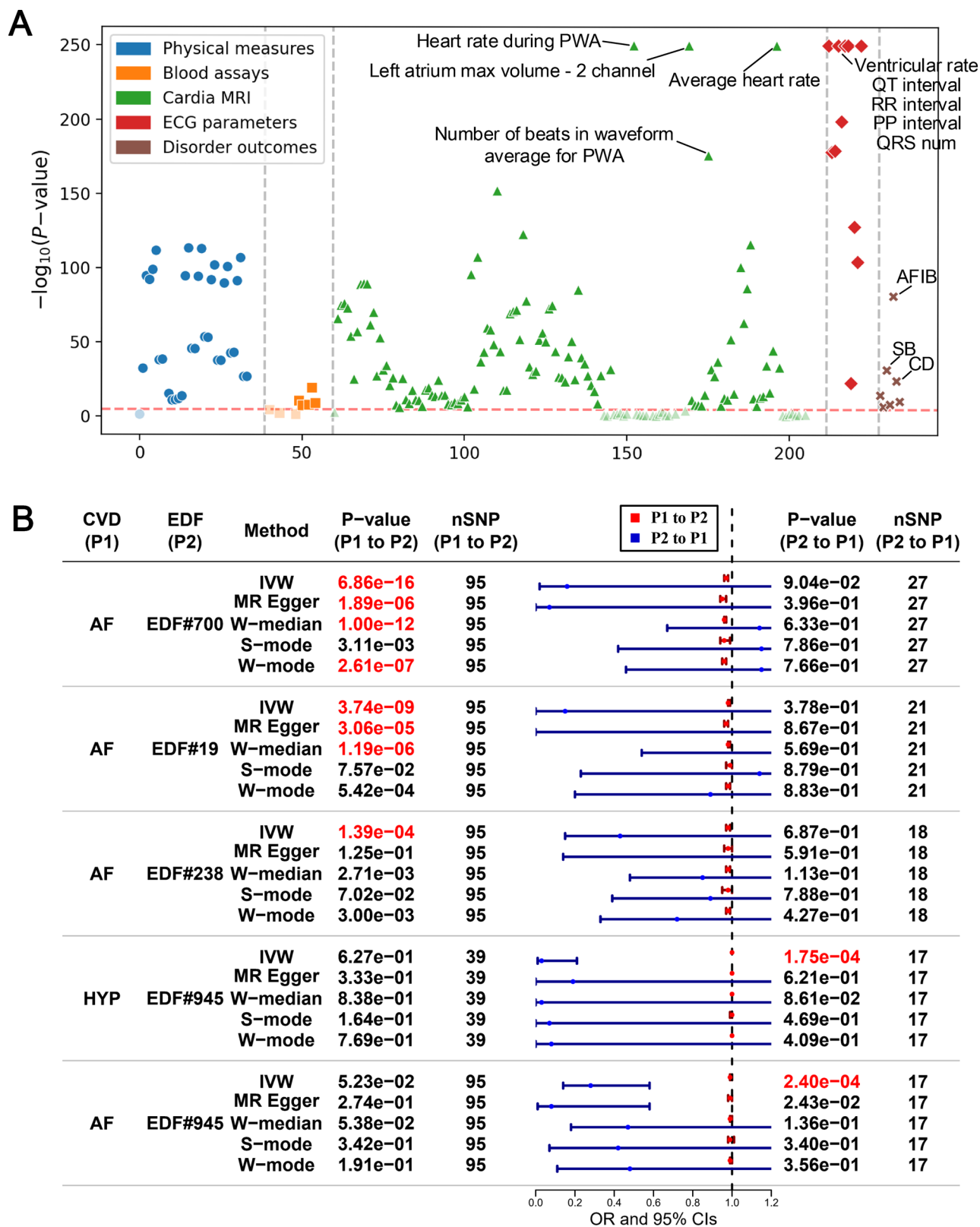


**Fig. 4 | Mapping important ECG-LFM-derived features to ECG regions.** Totally, 16 EDFs are considered as important for predicting CVDs. These EDFs are mapped to ECG regions by the perturbation-based EDF interpretation (PEI) method. The color bar is to show the absolute value of the  $t$  value (ranging from 0 to 1) in the  $t$ -test evaluating the difference between the ECG reconstructed by the perturbed EDF

and the ECG reconstructed by the original EDF. The ECG signals in blue are from healthy controls, and in pink are from the patients. MI myocardial infarction, HYP hypertrophy, CD conduction disturbance, STTC ST/T change, AFIB atrial fibrillation, GSVT general supraventricular tachycardia, SB sinus bradycardia, VE ventricular ectopics.

AUROC of 0.708) by 23.2% (Fig. 3D). With the increase of sample size in the training set, the performance of ECG-LFM is increased and is always better than others (Fig. 3D). All these models are further independently tested by the external set CODE-15. ECG-LFM fine-tuned by different sizes of the training set continued to exhibit the best performance with AUROC ranging from 0.833 to 0.915 compared to other methods (Fig. 3D).

**ECG-LFM-derived features interpret the associations between ECG morphology and CVDs**  
In predicting CVDs, ECG-LFM was pre-trained through self-supervised learning to capture features of ECGs by converting ECGs to 1024-dimensional embeddings (ECG-LFM-derived features, EDFs). For each individual, the importance score of each EDF was calculated according to the performance of the EDF in predicting CVDs for individuals from



**Fig. 5 | Exploring the phenotypic and genetic associations of important EDFs with CVDs and risk factors through PheWAS and Mendelian randomization analysis.** **A** PheWAS has identified phenotypic associations of 215 phenotypes with EDF. These phenotypes are 34 physical measures, 16 blood assays, 147 cardiac magnetic resonance imaging (CMRI) traits, 11 ECG traits, and 7 disorder outcomes. *P* values were derived from two-sided regression models (linear or logistic regression, depending on the trait type) as implemented in PHESANT. The red dotted lines indicate the Bonferroni-corrected significance threshold (*P* value =

$0.05/(16 \times 215) \approx 1.5 \times 10^{-5}$ ) showing significance in PheWAS. The lowest *P* value over the 16 EDFs in PheWAS is plotted. Extreme values (*P* value  $< 1 \times 10^{-250}$ ) were truncated at  $1 \times 10^{-250}$  to maintain axis scales. **B** Forest plot showing the putative causal relationships of EDFs with atrial fibrillation (AFIB) and hypertrophy (HYP). *P* values were derived from the inverse-variance weighted (IVW) method. Odds Ratios are presented as mean values, with error bars indicating 95% confidence intervals (CIs).

the PTB-XL and Chapman test sets by integrated gradient (IG)<sup>14</sup>. IG discovered 16 important EDFs (average important score  $>5.4 \times 10^{-3}$  across 2000 samples in the PTB-XL and Chapman test sets) in predicting eight types of CVDs (Supplementary Fig. S7).

To map these important EDFs to ECG morphology, we designed a PEI method. Using this method, the EDF of interest was perturbed and used to reconstruct ECG signals. The differences between the reconstructed ECG signals and the original ECGs were evaluated by paired *t*-tests. Regions with higher absolute *t* values were identified as most likely to be represented by the EDFs, and the smoothed *t* value maps highlighted these regions (see “Methods”).

Figure 4 shows the ECG regions represented by the top three most important EDFs in predicting each type of CVD. The ECGs of cases and controls in Fig. 4 were randomly selected from the PTB-XL and Chapman datasets. Out of these EDFs, EDF#102 is most likely to represent the ST-segment of ECG on V4 lead and is the key feature for predicting risks of myocardial infarction (MI) (importance score =  $5.4 \times 10^{-3}$ ) and ST/T change (STTC) (importance score =  $6.3 \times 10^{-3}$ ). Consistently, patients with MI or STTC exhibit abnormalities in the ST-segment, including ST-segment elevation or depression that reflects underlying ischemic or infarction processes<sup>15</sup> (Fig. 4). The EDF#19 is most likely to represent P-wave duration and P-wave amplitude on V2 lead, and is an important feature for predicting atrial fibrillation (AFIB). Indeed, P-wave duration and P-wave amplitude are reported as associated with AF by previous studies<sup>16,17</sup>.

Of the important EDFs, EDF#945 is most likely to represent the features of QRS morphology, particularly increased R wave amplitude on V5 lead (Fig. 4), and other features. EDF#238 and EDF#381 are most likely to represent S wave depth on V3 lead (Fig. 4). These three EDFs are the key features for predicting hypertrophy (HYP). A previous study has demonstrated that these QRS morphological changes reflect the increased ventricular mass and altered electrical conduction patterns due to hypertrophy<sup>18</sup>.

### Phenotypic associations between the important ECG-LFM-derived features and risk factors of cardiovascular disease

To identify the associations between 16 important EDFs and 215 CVD-related phenotypes (Supplementary Data 5), a phenome-wide association (PheWAS) study was performed by using information on 41,235 individuals from the UKB dataset. A total of 1387 significant associations were identified (Supplementary Data 6, *P* value  $< 0.05 / (16 \times 215) \approx 1.5 \times 10^{-5}$ ) between these EDFs and 167 CVD-related phenotypes that are 32 physical measures, 5 blood assays, 112 cardiac magnetic resonance imaging (CMRI) traits, 11 ECG traits, and 7 disorder outcomes (Supplementary Data 6). The associations of EDFs with the CMRI traits and ECG traits are more significant than with other phenotypes (Fig. 5A). Specifically, the EDFs are more likely to be associated with the ECG traits, including ventricular rate ( $\beta = -0.321$ , *P* value  $< 1.0 \times 10^{-250}$ ), QT interval ( $\beta = 0.241$ , *P* value  $< 1.0 \times 10^{-250}$ ), RR interval ( $\beta = 0.321$ , *P* value  $< 1.0 \times 10^{-250}$ ), PP interval ( $\beta = 0.337$ , *P* value  $< 1.0 \times 10^{-250}$ ) and QRS num ( $\beta = -0.488$ , *P* value  $< 1.0 \times 10^{-250}$ ), indicating the electrophysiological linkages between EDFs and fundamental cardiac conduction properties. Moreover, the EDFs are significantly associated with CMRI traits, including left atrium max volume - 2 channel ( $\beta = -0.131$ , *P* value  $< 1.0 \times 10^{-250}$ ), heart rate during PWA ( $\beta = -0.205$ , *P* value  $< 1.0 \times 10^{-250}$ ), number of beats in waveform average for PWA ( $\beta = 0.152$ , *P* value =  $1.0 \times 10^{-176}$ ), and LV circumferential strain AHA 9 ( $\beta = -0.139$ , *P* value =  $6.7 \times 10^{-153}$ ). This result indicates that EDFs can reflect underlying cardiac structure and function as measured by imaging techniques.

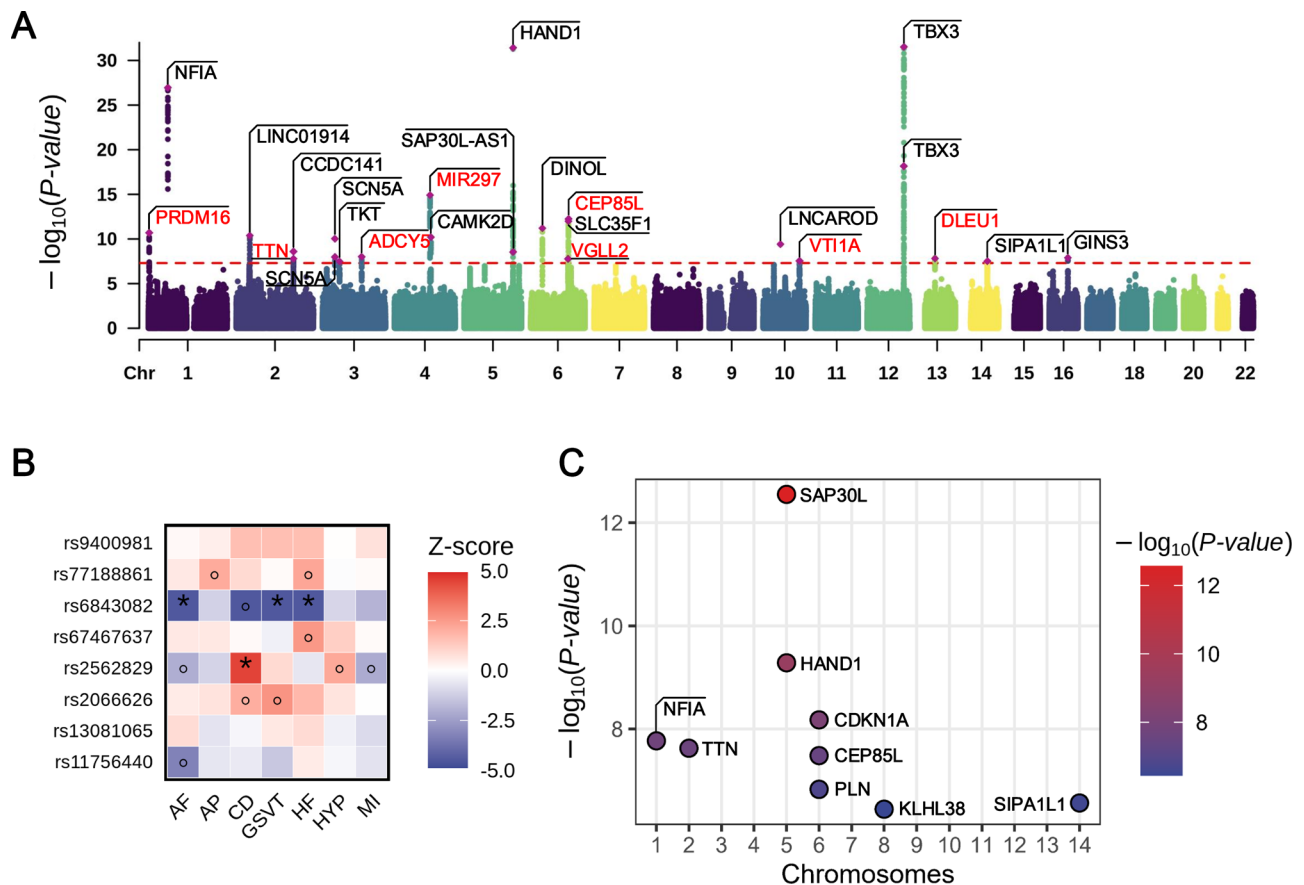
Interestingly, the EDFs are found significantly associated with the phenotypes of blood assays that are Cholesterol ( $\beta = 0.049$ , *P* value =  $2.7 \times 10^{-20}$ ), Sodium in urine ( $\beta = -0.035$ , *P* value =  $1.4 \times 10^{-11}$ ), Alanine aminotransferase ( $\beta = -0.031$ , *P* value =  $5.3 \times 10^{-10}$ ), and Alkaline phosphatase ( $\beta = -0.030$ , *P* value =  $9.6 \times 10^{-9}$ ). Many physical

measures are also identified as significantly associated with the EDFs, which include leg fat mass (right) ( $\beta = -0.095$ , *P* value =  $4.3 \times 10^{-114}$ ), leg fat mass (left) ( $\beta = -0.094$ , *P* value =  $1.2 \times 10^{-113}$ ), whole body fat mass ( $\beta = -0.117$ , *P* value =  $2.0 \times 10^{-112}$ ), and trunk fat mass ( $\beta = -0.117$ , *P* value =  $1.3 \times 10^{-107}$ ). These findings have demonstrated that the EDFs have the potential to monitor the risk factors associated with CVDs.

### Genetic associations between ECG-LFM-derived features and CVDs

To further detect biological meanings of EDF, we performed GWAS for 16 important EDFs on 41,235 individuals from the UKB dataset. The Manhattan plots and QQ plots for all 16 EDFs were shown in Supplementary Fig. S8 and Supplementary Fig. S9, respectively. Totally, 12 significant SNPs (*P* value  $< 5 \times 10^{-8}$ ,  $r^2 < 0.01$  within a 250 kb window) were identified (Supplementary Fig. S8, Supplementary Data 7). These SNPs are adjacent to 7 genes: *PLEKHA3*, *HAND1*, *FAM53A*, *SCN5A*, *NFIA*, *MIR297*, and *TBX3*. Notably, several of the identified genes (*HAND1*, *SCN5A*, *NFIA*, and *TBX3*) have been reported to be associated with PR interval<sup>19</sup>, RR interval<sup>2</sup>, and QT interval<sup>20</sup>, consistent with the PheWAS analysis showing significant associations (*P* value  $< 1.0 \times 10^{-250}$ ) between EDF and ECG intervals. The lead SNP rs1896347 near *TBX3* is significantly associated with EDF#19 ( $\beta = -0.026$ , *P* value =  $9.3 \times 10^{-10}$ ), which was a key feature for predicting AF. The individuals carrying rs1896347 are known to have a significantly increased risk for AF ( $\beta = 0.0402$ , *P* value =  $2.3 \times 10^{-7}$ )<sup>21</sup>. Two SNPs, rs13185595 ( $\beta = -0.014$ , *P* value =  $1.7 \times 10^{-9}$ ) and rs6587924 ( $\beta = 0.013$ , *P* value =  $2.6 \times 10^{-9}$ ), are identified as significantly associated with EDF#900 that is critical for predicting SB and VE. The SNP rs6587924 has been known to be associated with QRS duration<sup>22</sup> and QT interval<sup>23</sup>, which is located near *NFIA*. This aligns with the well-known role of *NFIA* in ventricular cardiomyocyte development that regulates SB and VE<sup>24</sup>.

To investigate the causal relationships between these EDFs and CVDs, we performed Mendelian randomization (MR) analyses by leveraging GWAS summary statistics for EDFs and CVDs. The GWAS summary statistics for CVDs (MI, AF, CD, GSVT, and HYP) were downloaded from the FinnGen database, which was generated based on more than 175 thousand samples. For SB, VE, and STTC, there are no GWAS based on a large sample size, and they were not used in MR analysis. The instrumental variants (IVs) for MR analysis are those with *P* value  $< 5 \times 10^{-8}$  and  $r^2 < 0.01$  for CVD and *P* value  $< 1 \times 10^{-5}$  and  $r^2 < 0.01$  for EDF. If the number of IVs for the exposure, EDF, or CVDs is less than five, the MR analysis was not conducted. After the quality control, 16 EDFs and 5 CVDs participated in the MR analysis. A significant causal relationship was determined if the *P* value surpassed the Bonferroni-corrected significance level (i.e., *P* value  $< 0.05 / (5 \times 16 \times 2) \approx 3.1 \times 10^{-4}$ , considering two causal directions). The bidirectional MR analysis revealed significant causal relationships between EDFs and the CVDs (Fig. 5B). When using CVD as exposure and EDF as outcome, the analysis indicated AF as a putative causal effect on the decreased EDF#700 ( $P_{IVW} = 6.9 \times 10^{-16}$ , OR = 0.968, 95% CI of 0.961–0.976), EDF#19 ( $P_{IVW} = 3.7 \times 10^{-9}$ , OR = 0.983, 95% CI of 0.978–0.989) and EDF#238 ( $P_{IVW} = 1.4 \times 10^{-4}$ , OR = 0.982, 95% CI of 0.973–0.992) (Fig. 5B, Supplementary Data 9). No horizontal pleiotropy was found by using the global test of MR-PRESSO. EDF#700, EDF#19, and EDF#238 are most likely to represent the ECG traits, R-wave and T-wave on I lead, P-wave on V2 lead and S-wave on V3 lead, respectively (Fig. 4). When using EDF as exposure and CVD as outcome, the MR analysis indicated that EDF#945 has putative causal effects on HYP ( $P_{IVW} = 1.8 \times 10^{-4}$ , OR = 0.033, 95% CI of 0.005–0.213) and AF ( $P_{IVW} = 2.4 \times 10^{-4}$ , OR = 0.285, 95% CI of 0.140–0.580). EDF#945 is most likely to present the features of QRS amplitude. This supports the notion that abnormality of QRS amplitude is a core feature of the ventricular depolarization associated with AF and HYP, consistent with previous pathological studies<sup>25,26</sup>.



**Fig. 6 | Genome-wide meta-analysis and gene association analysis for the first 10 principal components of 1024 EDFs.** **A** Manhattan plots showing the significance of SNPs identified by the meta-GWAS for the first 10 principal components of 1024 EDFs. The red dotted lines indicate the genome-wide significance threshold of  $P$  value =  $5 \times 10^{-8}$ . Each diamond indicates the nearest gene of the lead SNP. The nearest genes of the novel SNP are colored in red.  $P$  values were obtained from two-sided meta-analysis using METAL, based on linear mixed models computed by BOLT-LMM. **B** Heatmap to show the correlation between the identified novel SNPs and CVDs. The darker colors represent higher z-scores (beta/se) obtained from the published GWAS summary statistics. “\*” denotes SNP with significance of  $P$

value  $< 5 \times 10^{-8}$  in GWAS for EDFs. “o” denotes the SNP with significance of  $P$  value  $< 0.05$  in GWAS for EDFs. Z-scores were clipped to the range  $[-5, 5]$  for visualization, with values exceeding this threshold set to the nearest bound (i.e.,  $>5$  as 5 and  $<-5$  as  $-5$ ). **C** Heatmap to show the significance of genes identified by MAGMA analysis for EDFs.  $P$  values were calculated using the two-sided multiple linear principal components regression model in MAGMA. The horizontal axis represents the negative  $\log_{10}$ -transformed  $P$  value, with significance defined by the Bonferroni-corrected threshold ( $P$  value  $< 0.05/18762 \approx 2.7 \times 10^{-6}$ ). Increasingly red hues represent higher significance.

### Empowering genetic discovery for ECG by using ECG-LFM-derived phenotypes

Most of the GWAS for ECG are performed only for the known risk factors associated with CVDs because of the difficulty in reducing the data dimension and keeping the biological meaning of the ECG signal. To expand genetic insights into ECG, we extracted the first 10 principal components (PCs) from 1024 EDFs and performed GWAS for each PC on 41,235 individuals of the UKB dataset. These 10 PCs accounted for 85.2% of the total variance in 1024 EDFs, suggesting they capture the majority of the electrocardiographic variability of the individuals. For 10 PCs of the EDFs, a multi-trait genome-wide meta-analysis was performed using METAL<sup>27</sup>. We identified 24 significant SNPs ( $P$  value  $< 5 \times 10^{-8}$ ,  $r^2 < 0.01$  within a 250 kb window) at 19 loci from the meta-GWAS (Fig. 6A, Supplementary Data 10). Of the 24 significant SNPs, 16 were located  $< 100$  kb away from the lead SNPs reported by the most recently published GWAS based on more than 77,898 samples for the ECG traits, including spatial QRS-T angle, PR interval, QT interval, P wave duration and QRS duration (Supplementary Fig. S10 and Supplementary Data 11)<sup>28–30</sup>. The remaining 8 significant SNPs (rs77188861, rs2562829, rs13081065, rs6843082, rs9400981, rs11756440, rs67467637 and rs2066626) are novel findings of this study. Among these 8 novel SNPs, 5 are intronic variants located within

*CEP85L*, *PRDM16*, *ADCY8*, *DLEU1*, and *VT11A*, two are intergenic variants close to genes, *VGLL2* and *MIR297*, and one is an exonic variant located within *TTN* (Supplementary Fig. S10). Indeed, 6 of the 8 novel SNPs are reported to be significantly associated with CVDs<sup>21,31,32</sup> (Fig. 6B, Supplementary Data 12 and 13). Especially, the variant rs6843082 ( $P$  value =  $1.2 \times 10^{-15}$ ), located near *MIR297*, is reported to be significantly associated with atrial fibrillation (beta =  $-0.371$ ,  $P$  value =  $3.4 \times 10^{-155}$ )<sup>21</sup>, generalized supraventricular tachycardia (beta =  $-0.100$ ,  $P$  value =  $1.3 \times 10^{-8}$ )<sup>32</sup>, and heart failure (beta =  $-0.075$ ,  $P$  value =  $2.6 \times 10^{-15}$ )<sup>31</sup>, indicating its pleiotropic effects on cardiac conduction and ventricular function (Fig. 6B).

To identify genes associated with the EDFs, multi-marker analysis of GenoMic annotation (MAGMA)<sup>33</sup> and FUMA<sup>34</sup> on the meta-GWAS result was performed. In total, 63 unique genes were identified by at least one of the three FUMA gene mapping strategies or considered as significant by MAGMA. Briefly, MAGMA analysis revealed 9 genes (*NFIA*, *TTN*, *SAP30L*, *HAND1*, *CDKN1A*, *CEP85L*, *PLN*, *KLHL38*, and *SIPA1L1*) as significantly ( $P$  value  $< 0.05/18762 \approx 2.7 \times 10^{-6}$ ) associated with the EDFs (Fig. 6C, Supplementary Data 14). FUMA identified a total of 62 candidate genes (Supplementary Data 15). Out of these 63 genes, 39 are reported as associated with ECG traits, such as QRS duration<sup>35</sup>, QT interval<sup>36</sup>, and PR interval<sup>28</sup>. Moreover, 28 of them were found by

previous studies to be associated with CVDs in the GWAS Catalog<sup>37</sup>, such as myocardial infarction<sup>38</sup>, heart failure<sup>31</sup>, and atrial fibrillation<sup>21</sup>. The MAGMA analysis indicated *SAP30L* as the most significantly associated with the EDFs ( $P$  value =  $2.8 \times 10^{-13}$ ). *SAP30L* is known to play roles in the regulation of cardiac development and heart function<sup>39</sup>. These findings suggest that the genes identified in our analysis may be involved in the genetic architecture of ECG features and CVDs.

## Discussion

In this study, we introduced an ECG foundation model, ECG-LFM, and evaluated its generalizability in adapting to diverse downstream tasks. ECG-LFM significantly outperformed other methods for predicting various CVDs across multiple datasets, including PTB-XL, Chapman, CODE-15, CPSC-2018, and UK Biobank, with an average AUROC of 0.930 (ranging from 0.807 to 0.993). These results highlight the ability of ECG-LFM to be applied across diverse populations and clinical scenarios. The better performance of ECG-LFM is attributed to its self-supervised pre-training on over ten million ECGs, which enabled it to capture intricate spatiotemporal patterns in ECG signals. Unlike traditional rule-based systems or task-specific deep learning models, ECG-LFM leverages a multi-task learning strategy that combines contextual contrastive loss, MLM loss, and multi-segment contrastive loss, enabling the model to align local ECG features with their contextual representations, reconstruct masked segments to learn long-range relationships, and enforce patient-level consistency across ECG segments, respectively.

The generalizability of ECG-LFM was evaluated across multiple dimensions. First, we tested the model on independent datasets with diverse demographic characteristics: PTB-XL ( $N=18,118$ , European ancestry), Chapman ( $N=10,646$ , Chinese ancestry), CPSC-2018 ( $N=6877$ , Chinese ancestry), and UK Biobank ( $N=41,235$ , mainly European ancestry). These datasets represent populations from different geographic regions and healthcare systems. Second, the datasets include individuals with varying clinical presentations, ranging from healthy controls to patients with acute myocardial infarction, arrhythmias, and conduction disorders. Third, the model was evaluated on ECGs acquired using different recording devices and protocols, demonstrating robustness to technical variations. Detailed information for these datasets was shown in Supplementary Data 1 and Supplementary Fig. S1. The consistent performance of ECG-LFM across these diverse settings (average AUROC of 0.930, ranging from 0.807 to 0.993) supports its generalizability to different populations and clinical scenarios. The advantage of ECG-LFM was also particularly evident in scenarios with limited labeled data. For instance, when fine-tuned on just 1% of the PTB-XL dataset, ECG-LFM outperformed other models by up to 23.2% in AUROC. This finding is clinically significant, as it addresses one of the major challenges in medical AI, the scarcity of high-quality labeled data. By minimizing the need for extensive annotated datasets, ECG-LFM enables quick implementation even in resource-constrained environments.

One of the critical barriers to the adoption of AI in clinical practice is the “black-box” nature of many deep learning models. ECG-LFM addresses this challenge through saliency maps generated using Grad-CAM. These maps provide visual explanations of the predictions, highlighting regions of the ECG that contribute most significantly to the diagnosis. For example, in cases of myocardial infarction (MI) and conduction disturbances (CD), the saliency maps of ECG-LFM aligned closely with expert annotations (Supplementary Fig. S6), while also revealing subtle patterns that might be overlooked during manual analysis. This transparency not only enhances clinician trust but also facilitates the integration of AI into routine diagnostic workflows. Furthermore, the PEI method allowed us to link specific ECG-LFM-derived features to clinically relevant ECG waveforms. For instance, EDF#102 was associated with ST-segment abnormalities in lead V4, a well-known marker of MI<sup>40</sup>. For the prediction of hypertrophy (HYP), EDF#945 and

EDF#238 are considered as the most significant morphological features. These two features are most likely to represent QRS morphology according to the PEI method. Indeed, QRS morphological changes have been discovered to reflect the increased ventricular mass and altered electrical conduction patterns due to hypertrophy<sup>18</sup>. Such interpretability is crucial for validating the predictions and ensuring their alignment with established cardiological knowledge.

A key contribution of this study is on using EDFs to uncover genetic associations underlying ECG morphology. Traditional GWAS rely on handcrafted ECG features (e.g., QT interval), which are limited in their ability to capture complex waveform interactions. In contrast to traditional models that necessitate high-quality labeled datasets for downstream analysis, ECG-LFM leverages self-supervised pre-training to extract comprehensive, label-independent representations, referred to as ECG-LFM-derived features (EDFs). While previous studies have linked deep learning-derived features to genetic variants<sup>41</sup>, our approach uniquely combines deep representations from an ECG foundation model with comprehensive genetic and phenotypic analyses to provide novel insights into the genetic architecture of cardiac electrophysiology. ECG-LFM enables the discovery of novel genotype-phenotype relationships, paving the way for personalized risk assessment and targeted interventions. Through GWAS for EDFs, we identified 24 independent SNPs at 19 loci where genetic associations reached the genome-wide significance threshold ( $P$  value  $< 5 \times 10^{-8}$ ,  $r^2 < 0.01$  within a 250 kb window), including 8 new SNPs located within *CEP85L*, *PRDM16*, *ADCY5*, *DLEU1*, *VTG1A*, *VGLL2*, *MIR297*, and *TTN*. These findings provide new insights into the genetic architecture of cardiac electrophysiology and its relationship with CVDs. Among these novel SNPs, the lead SNP rs2562829 is near *TTN*, which is a gene known to be associated with QT interval<sup>42</sup> and PR interval<sup>43</sup>. While *TTN* is an established gene for arrhythmia<sup>44</sup>, the specific variant rs2562829 represents a novel association with cardiac conduction. Another intergenic variant, rs6843082, is significantly associated with EDF and is known to be significantly associated ( $P$  value  $< 5 \times 10^{-8}$ ) with AF<sup>21</sup>, generalized supraventricular tachycardia (GSVT)<sup>32</sup>, and heart failure (HF)<sup>31</sup>. The ability of ECG-LFM to derive phenotypes that correlate with genetic variants represents a paradigm shift in electrogenomic research.

This study has several limitations. First, the limited availability of accessible datasets containing both ECG and CVD information restricts our comprehensive evaluation of ECG-LFM. Expanding available datasets is crucial to propel research forward in this challenging domain. Second, the computational resources required for pre-training and fine-tuning ECG-LFM are substantial. Although the efficiency of ECG-LFM improves with transfer learning, real-time deployment in clinical settings may still pose challenges. Third, while the interpretability of ECG-LFM is a step forward, further validation is needed to ensure that the explanations align with clinical reasoning across a broader range of conditions. Collaborative efforts with cardiologists to refine the saliency maps and EDF interpretations will be essential.

In conclusion, ECG-LFM represents a transformative approach to ECG analysis, combining state-of-the-art performance, interpretability, and genetic discovery. By leveraging self-supervised learning on large-scale datasets, the model overcomes many of the limitations of traditional methods, offering a robust and generalizable framework for CVD diagnosis and prediction. ECG-LFM further provides new insights into the genetic architecture of cardiac electrophysiology and its relationship with CVDs. With continued refinement and validation, ECG-LFM has the potential to revolutionize the way we diagnose, monitor, and treat CVDs.

## Methods

### Ethics

The UK Biobank has approval from the North West Multi-Centre Research Ethics Committee as a Research Tissue Bank (application ID: 51732).

## Data sources

In this study, we collected a total of 11,571,587 12-lead ECG records of 1,979,599 individuals in HEEDB<sup>10</sup> and MIMIC-IV-ECG dataset<sup>11</sup> for model pre-training. The HEEDB dataset is currently the largest open-access ECG dataset, containing 10,608,417 ECG records of 1,818,247 individuals. The MIMIC-IV-ECG dataset is part of the MIMIC series, containing 800,035 ECG records of 161,352 individuals treated in the intensive care unit (ICU). Most of the 12-lead ECG records in HEEDB and MIMIC-IV-ECG are 10 s long with a sampling rate of 500 Hz.

Our task-specific data comprised five ECG datasets: PTB-XL<sup>45</sup>, Chapman<sup>46</sup>, CODE-15<sup>47</sup>, CPSC-2018<sup>48</sup>, and UK Biobank<sup>49</sup>. The PTB-XL dataset contains 21,837 12-lead ECG records of 18,885 individuals, where each ECG signal is 10 s long with a sampling rate of 500 Hz. It also contains records of various CVDs grouped into five super-diagnostic classes: conduction disturbance (CD), hypertrophy (HYP), myocardial infarction (MI), ST/T change (STTC), and ventricular ectopics (VE). A detailed summary of the size of the individual classes is presented in Supplementary Fig. S1 and Supplementary Data 1.

The Chapman dataset consists of 10,646 12-lead ECG records of 10,646 individuals. Each ECG signal is 10 s long and sampled at a frequency of 500 Hz. The Chapman dataset contains six types of CVDs that are atrial flutter (AF), atrial fibrillation (AFIB), sinus irregularity (SI), sinus bradycardia (SB), sinus tachycardia (ST), and supraventricular tachycardia (SVT). These six types of CVDs were grouped into three classes following a previous study<sup>46</sup>, which are sinus bradycardia (SB), atrial fibrillation (AFIB), and general supraventricular tachycardia (GSVT) (Supplementary Fig. S1, Supplementary Data 1).

CODE-15 is an ECG database that includes 345,779 ECG records collected from 233,770 patients. The ECG records in this database were pre-processed as in a previous study<sup>50</sup>. CODE-15 contains patient diagnosis information for five types of CVDs that are atrial fibrillation (AFIB), sinus bradycardia (SB), sinus tachycardia (ST), left bundle branch block (LBBB), and right bundle branch block (RBBB). We grouped these CVDs into their upper-level types, resulting in four super-diagnostic classes that are sinus bradycardia (SB), atrial fibrillation (AFIB), general supraventricular tachycardia (GSVT), and conduction disturbance (CD) (Supplementary Fig. S1, Supplementary Data 1).

The CPSC-2018 dataset consists of 9569 12-lead ECG records of 6877 individuals. The ECG is sampled at a rate of 500 Hz. It contains patient diagnosis information for seven types of CVDs (Supplementary Fig. S1, Supplementary Data 1), which are atrial fibrillation (AFIB), first-degree atrioventricular block (I-AVB), left bundle branch block (LBBB), right bundle branch block (RBBB), ST-segment depression (STD), ST-segment elevation (STE), and ventricular ectopics (VE). Following a previous study<sup>51</sup>, we grouped the CVDs into four upper-level types, resulting in conduction disturbance (CD), ST/T change (STTC), atrial fibrillation (AFIB), and ventricular ectopics (VE), where CD consists of I-AVB, LBBB, and RBBB and STTC consists of STD and STE.

The UK Biobank (UKB) is a longitudinal study of more than 500,000 participants aged from 40 to 69. At baseline assessment, participants provided genotype data, information on health and lifestyle via questionnaire, physical measurement, and donated samples of blood and urine. A subgroup of participants was invited back for subsequent visits for additional investigations, including collecting an ECG. Totally, 41,235 participants in the UKB have both ECG data and genotype data. The ECG signal of each individual in the UK Biobank was a 12-lead and 10-s-long signal with a sampling rate of 500 Hz. ICD-9 and ICD-10 codes were used to define the presence or absence of CVDs for each individual. There are patients diagnosed as conduction disturbance (CD), hypertrophy (HYP), myocardial infarction (MI), ST/T change (STTC), ventricular ectopics (VE), sinus bradycardia (SB), atrial fibrillation (AFIB), and general supraventricular tachycardia (GSVT) (Supplementary Fig. S1, Supplementary Data 1). The UKB has approval

from the North West Multi-Centre Research Ethics Committee as a Research Tissue Bank (Application ID: 51732).

## ECG data pre-processing

To standardize ECG data from multiple sources, we applied linear interpolation to sample the data to force the sampling frequency to be 500 Hz. To address signal artifacts, we implemented a series of filtering steps. First, a high-pass filter (1 Hz cutoff) was applied to remove baseline drift, followed by a second-order Butterworth low-pass filter (30 Hz cutoff) to attenuate high-frequency noise, and finally, a notch filter (50/60 Hz) was employed to suppress powerline interference. If the duration of the ECG was less than 10 s (5000 data points), zero padding was used to extend the ECG signal into 10 s for each lead. All signals were normalized using z-score standardization and segmented into non-overlapping 5 s segments before inputting them into the model. The ECG data pre-processing pipeline is shown in Supplementary Fig. S12.

## Constructing the electrocardiogram evaluation foundation model

The electrocardiogram foundation model, ECG-LFM, is composed of a feature encoder that extracts latent representations (tokens) from raw 12-lead ECG inputs. The feature encoder consists of several blocks containing a 1D-convolution layer followed by layer normalization and a GELU activation function<sup>52</sup>. It takes a raw 12-lead ECG as input and outputs latent representations  $z_1, z_2, \dots, z_T$  for  $T$  time-steps.

A quantization module is included in ECG-LFM for discretizing the latent representations to the target context vector and the discretized ID. The quantization module is used to discretize the latent representations  $z_t$  of the feature encoder to a finite set of representations. This module selects discrete codebook entries from multiple codebooks<sup>53</sup> and concatenates them, which has resulted in an efficient and compact representation of encoded features. To enable differentiable selection of discrete codebook entries, the Gumbel softmax is employed, which allows the quantization process to be integrated into the end-to-end training pipeline. Specifically, the quantization module utilizes  $G$  codebooks, each of which contains  $V$  entries  $e \in \mathbb{R}^{V \times d/G}$ , where  $d$  is the dimensionality of the feature encoder output  $z_t$ . The feature encoder output  $z_t$  is mapped to logits  $l \in \mathbb{R}^{V \times G}$ , which represent the scores for each entry in the codebooks. For each codebook, one entry  $e_t$  is selected according to the probability  $p_{g,v}$ , which is computed as

$$p_{g,v} = \frac{\exp((l_{g,v} + n_v)/\tau)}{\sum_{k=1}^V \exp((l_{g,k} + n_k)/\tau)} \quad (1)$$

where  $\tau$  is a non-negative temperature parameter controlling the sharpness of the distribution,  $n = -\log(-\log(u))$  and  $u$  are uniform samples from uniform distribution  $\mathcal{U}(0,1)$ . Finally, the resulting vectors  $e_1, e_2, \dots, e_G$  are concatenated to form the discretized ID for each token, and the linear transformation is applied to obtain target context vectors  $q_t \in \mathbb{R}^f$  for each latent representation  $z_t$ .

ECG-LFM used an embedding module for extracting contextual representations for each token, as well as three tasks that are a contextual contrastive task to identify the quantized representation from the quantization module, an MLM task for predicting the discretized ID of each token and a multi-segment contrastive task to capture functionally relevant information (Fig. 1A). The embedding model of ECG-LFM is inspired by BERT<sup>54</sup>. ECG-LFM masks the token  $z_t$  from the feature encoder and reconstructs the masked tokens by leveraging contextual information from the surrounding tokens. Each token has a 6.5% probability of being a starting index, where if selected, we mask the subsequent 10 tokens. This results in approximately 49% of sample tokens being masked. The masked tokens were then fed into the linear

projection layer, followed by the stack of conformer blocks<sup>55</sup> to produce high-level contextual vectors  $c_t$ . Conformer is a hybrid neural network architecture that integrates depth-wise convolution (enabling multi-scale feature extraction) with multi-headed self-attention mechanisms (capturing long-range dependencies) to enhance the modeling of both local acoustic patterns and global contextual relationships in ECG signals.

### Contextual contrastive loss

In order to align the target context vectors (from the quantization module) with predicted contextual representations (from the embedding module), a contrastive loss is used to identify the true quantized latent representation for a masked time step within a set of distractors. For each masked token, we maximize the cosine similarity between the contextual representation  $c_t$  and the target context vectors  $q_t$  in a set of  $K+1$  quantized candidate representations  $Q_t$ . The loss is defined as:

$$\mathcal{L}_{CC} = -\log \frac{\exp(\text{sim}(c_t, q_t)/K)}{\sum_{\tilde{q} \in Q_t} \exp(\text{sim}(c_t, \tilde{q})/K)} \quad (2)$$

### Masked language modeling (MLM) loss

For the output vector of each token from the final conformer block of the embedding module, the softmax layer takes the vector as input and predicts its associated token ID, which was previously assigned by the quantization module. We represent the cross-entropy loss for this masked prediction task as:

$$\mathcal{L}_{MLM} = -\sum_{k=1}^V y_k \log(p_k) \quad (3)$$

where  $V$  is the size of the codebook, and  $p_k$  is the predicted probability for the  $k$ -th entry.

### Multi-segment contrastive loss

In order to capture functionally relevant information in the ECG signal, we employed multi-segment contrastive loss<sup>56</sup> to help ECG-LFM learning consistent representations between consecutive ECG segments. Specifically, given an ECG signal  $x_i$  with duration  $S$  seconds from patient  $i$ , we can extract  $N$  non-overlapping ECG segments  $x_1^i, x_2^i, \dots, x_N^i$ , each with duration  $S/N$ . We maximize the cosine similarity between the feature representations of segments from the same patient while minimizing the cosine similarity between segments from different patients. This is achieved through a contrastive loss function that operates on both forward and reverse positive pairs. We used  $N=2$  in this study, and obtained segments  $x_A^i$  and  $x_B^i$  from patient  $i$ . The loss is defined as:

$$\mathcal{L}_{MC} = \frac{1}{2}(\mathcal{L}_{MC_{for}} + \mathcal{L}_{MC_{rev}}) \quad (4)$$

$$\mathcal{L}_{MC_{for}} = -\log \frac{\exp(\text{sim}(f_\theta(x_A^i), f_\theta(x_B^i))/\tau)}{\sum_{k=1}^K \exp(\text{sim}(f_\theta(x_A^i), f_\theta(x_B^k))/\tau)} \quad (5)$$

$$\mathcal{L}_{MC_{rev}} = -\log \frac{\exp(\text{sim}(f_\theta(x_B^i), f_\theta(x_A^i))/\tau)}{\sum_{r=1}^R \exp(\text{sim}(f_\theta(x_B^i), f_\theta(x_A^r))/\tau)} \quad (6)$$

where  $f_\theta$  is the feature encoder,  $\text{sim}(\cdot)$  represents the cosine similarity,  $\tau$  is a non-negative temperature parameter, and  $r$  is the index of negative segments.

The final training loss to be minimized is:

$$\mathcal{L} = \alpha \mathcal{L}_{CC} + \beta \mathcal{L}_{MLM} + \gamma \mathcal{L}_{MC} \quad (7)$$

where  $\alpha$ ,  $\beta$ , and  $\gamma$  are set to 1,  $\mathcal{L}_{CC}$  is contextual contrastive loss, and  $\mathcal{L}_{MLM}$  is MLM loss.

### Model training

ECG-LFM was pre-trained on 11,571,587 ECGs using two A800 120GB GPUs with distributed data parallelism and gradient accumulation. In the pre-training process, we employed the Adam optimizer<sup>57</sup> with a batch size of 64, Adam eps of  $1 \times 10^{-6}$ , learning rate of  $5 \times 10^{-5}$ ,  $\beta_1$  of 0.9, and  $\beta_2$  of 0.98. The training was halted after 100 epochs. For model fine-tuning, we randomly partitioned the training data into a training set (80%) and a validation set (20%). ECG-LFM was optimized by running on one A800 120GB GPU. Checkpoints were selected on the respective validation set using early stopping conditioned on validation loss. Early stopping is configured to halt training if the validation loss does not improve within 10 epochs. Models were trained with batch size 256 for a maximum of 100 epochs using the Adam optimizer<sup>57</sup> with Adam eps of  $1 \times 10^{-8}$ , learning rate of  $1 \times 10^{-6}$ ,  $\beta_1$  of 0.9, and  $\beta_2$  of 0.98 in each epoch using an exponential scheduler. For the individual recognition task, we extracted the embedding from a pre-trained model and fed it into a single-layer perceptron for prediction. The single-layer perceptron was optimized using the Adam algorithm with backpropagation, trained for 500 epochs with a batch size of 512, Adam eps of  $1 \times 10^{-8}$ , learning rate of  $1 \times 10^{-3}$ ,  $\beta_1$  of 0.9, and  $\beta_2$  of 0.99.

### Individual recognition using ECG-LFM-derived features

To evaluate the capability of ECG-LFM for individual recognition, we pre-processed the ECG records as described in the “ECG data pre-processing” section. For each individual, we randomly selected one 5-s segment to construct the test set, and all remaining segments constructed the training set. By using this strategy, from the Chapman dataset, 10,646 segments were used to construct the training set, and 10,646 segments were used to construct the test set. From the CPSC-2018 dataset, 13,286 segments were used to construct the training set, and 6876 segments were used to construct the test set. Each ECG segment was passed through the ECG-LFM to obtain the 1024-dimensional embedding, then we trained a single-layer perceptron to map the embedding to one of the  $N$  individual IDs via cross-entropy (CE) loss:

$$\mathcal{L}_{CE} = -\log \left( \frac{\exp(s_j)}{\sum_{j=1}^N \exp(s_j)} \right) \quad (8)$$

where  $s_j$  is the  $j$ -th logit (unnormalized score) output by the perceptron, and  $s_y$  is the logit corresponding to the individual ID  $y_i$ . During testing, we computed the scores (softmax normalized probabilities), and the predicted individual ID is considered correct if the true individual ID appears within the top 3 of the ranked scores. We evaluated the individual recognition performance using Top-3 Classification Accuracy (Top-3 ACC). The overall accuracy is calculated as:

$$\text{Top-3 ACC} = \frac{\sum_{i=1}^M \mathbb{I}(y_i \in \text{Top}_3(p_i))}{M} \quad (9)$$

where  $M$  is the total number of test samples,  $y_i$  is the true individual ID of the  $i$ -th test sample,  $p_i$  is the predicted normalized probability vector output by the model for the  $i$ -th sample.

### Developing a perturbation-based model to interpret ECG-LFM-derived features

To identify the ECG morphology represented by a specific ECG-LFM-derived feature (EDF), we develop a PEI method. First, a convolutional decoder is established to reconstruct ECGs by the EDFs in the pre-trained ECG-LFM. The EDF decoder consisted of four blocks followed by a final convolutional block. Each decoder block comprises two blocks of a 1D-convolution layer with a kernel size of 3, batch normalization, and LeakyReLU activation function, followed by a 1D transposed convolution operator with a stride of 2 and kernel size of 2. The final convolutional block comprises two blocks of a 1D-convolution layer with a kernel size of 3, batch normalization, and

LeakyReLU activation function, followed by a 1D-convolution layer with kernel size 1. To train and evaluate the ECG decoder, we randomly extracted 5000 samples, 1000 samples, and 2000 samples from the PTB-XL dataset to form the training set, validation set, and test set, respectively.

For each EDF, we added one standard deviation ( $\sigma$ ) as a perturbation. One perturbed EDF, together with other original EDFs, was input into the decoder model to reconstruct the ECG. The process was repeated for 200 randomly selected ECGs from the test set to improve the result's robustness. The reconstructed ECG by the perturbed EDFs was compared to the ECG reconstructed by the original EDFs using the paired  $t$ -test in 200 ECGs. The ECG region showing a higher absolute value of  $t$  value is considered to have high potential to be represented by the EDF. Using the  $t$  value, an absolute  $t$ -map of each EDF was constructed, and a Gaussian filter ( $\sigma = 3$ ) was used to smooth the  $t$ -map.

### Quality control of SNP for genome-wide association study (GWAS)

The quality control of the SNPs was performed for the genotype data of samples using PLINK 1.9<sup>58</sup>. SNPs were excluded if they had a minor allele frequency  $< 0.01$ , an imputation INFO score  $< 0.3$ , a Hardy-Weinberg equilibrium  $P$  value  $< 1 \times 10^{-6}$ , minor allele counts  $< 5$ , or call rate  $< 0.95$ . This process yielded a final set of 8.52 million SNPs for the GWAS.

### Genome-wide association study for ECG-LFM derived features

A GWAS was performed for each EDF of European ancestry individuals in the UK Biobank (UKB). The GWAS was carried out by BOLT-LMM<sup>59</sup>, which is a Bayesian-based linear mixed model (LMM) allowing for the relatedness of samples by adjusting the population structure and cryptic relatedness. BOLT-LMM was corrected for age, sex, genotype batch, assessment center, and the first 10 genetic principal components (PCs), and was fitted by random effects of the restricted SNPs (with  $LD^2 < 0.9$ ). The SNPs used in BOLT-LMM were calibrated by the LD scores calculated from the 1000 Genomes Project Europeans reference<sup>60</sup>. Genotyped SNPs were imputed according to the Haplotype Reference Consortium reference panel<sup>61</sup>.

### Mendelian randomization (MR) analysis

We performed bidirectional Mendelian randomization (MR) analyses to investigate causal relationships between EDFs and CVDs. The MR analysis is to test two causal directions: (1) EDF as exposure and CVD as outcome, and (2) CVD as exposure and EDF as outcome. When EDF is taken as exposure and CVD as outcome, the MR is to test whether genetic variants associated with EDFs causally influence CVD risk. When CVD is considered as exposure, and EDF is considered as outcome, the MR is to test whether genetic variants associated with CVDs causally influence EDF expression. To distinguish causality from horizontal pleiotropy, five MR models were applied, which are based on different assumptions on horizontal pleiotropy, including inverse-variance-weighted (IVW), MR-Egger, weighted mode, weighted median, and simple mode. IVW integrates the GWAS effect ratios of instrumental SNPs and uses a weighted linear regression to calculate the causal estimate without correcting horizontal pleiotropy<sup>62</sup>. MR-Egger is a simple modification of IVW through the addition of an intercept term that captures the non-zero (weighted) average pleiotropic effect<sup>63</sup>. The weighted mode model calculates the causal effect from the most frequent instrumental SNPs, which loosens the assumption of the weighted median model (i.e., pleiotropy occurs in less than half of the instrumental SNPs)<sup>64</sup>. The weighted median model measures the causal effect using the weighted median of the SNP ratio to reduce the impact from pleiotropic SNPs<sup>65</sup>. The simple mode method estimates the causal effect by selecting the most frequently occurring effect value among instrumental SNPs, assuming it best represents the true causal relationship while reducing bias from pleiotropy. The MR analyses were performed by the R package

"TwoSampleMR" (version 0.5.5). For these five MR methods, we selected instrumental SNPs with GWAS  $P$  value  $< 5 \times 10^{-8}$  for CVD and GWAS  $P$  value  $< 1 \times 10^{-5}$  for EDF, clumped with  $LD^2 < 0.01$  within a 1 Mb window using PLINK 1.9<sup>58</sup> according to the same 1000 Genomes Project reference described above<sup>60</sup>. We obtained 160 exposure-outcome pairs with the number of instrumental SNPs (SNPs robustly associated with the risk factor/exposure of interest) larger than 5. A significant causal relationship was determined if the  $P$  value surpassed the Bonferroni-corrected significance level (i.e.,  $0.05/160 \approx 3.1 \times 10^{-4}$ ). We further verified the significant MR results after correction by sensitivity analysis. Specifically, we conducted MR-PRESSO to detect the presence of horizontal pleiotropy ( $P$  value  $< 0.05$ ).

### Phenome-wide association study (PheWAS)

PheWAS is performed to examine the associations between CVD-related risk factors and the important EDFs by PHEASANT<sup>66</sup>. The model was adjusted for age, sex, genotype batch, assessment center, and the first 10 genetic principal components. The phenotypes considered in this study are mainly related to cardiovascular functions and disease risk factors, which include 34 physical measures, 16 blood assays, 147 CMRI traits, 11 ECG traits, and 7 disorder outcomes. All information is from the UK Biobank. In PheWAS analysis, if the  $P$  value is lower than  $0.05/(16 \times 215) \approx 1.5 \times 10^{-5}$ , the association between the EDF and other phenotypes is considered as significant. Here, the Bonferroni correction accounts for multiple tests for 16 EDFs and 215 cardiovascular biomarkers.

### Gene mapping using FUMA

To strengthen our gene-level findings, we have further performed genome-wide gene association analysis using FUMA<sup>34</sup> on the meta-GWAS result. Loci obtained by meta-GWAS were mapped to genes in FUMA using three strategies. First, SNPs were mapped to genes based on physical distance (within a 10-kb window) from known protein-coding genes in the human reference assembly (GRCh37/hg19). Secondly, SNPs were also mapped to genes if they have shown significant eQTL association with the genes. The eQTL mapping was performed using two data repositories (GTEx v8 and eQTL catalog), and was based on cis-eQTLs that can map SNPs to genes up to 1 megabase apart. We used a false discovery rate of 0.05 to define significant eQTL associations. Thirdly, SNPs were mapped to genes using the three-dimensional spatial interactions between SNPs and genes, which potentially involve long-range interactions. This procedure was performed using Hi-C data from the study by Schmitt et al.<sup>67</sup> in the FUMA platform. The candidate genes were selected if the SNP-gene interaction region overlaps with an enhancer region according to the data from Heart tissue in the Roadmap Epigenomics Project<sup>68</sup>, and is located in the promoter region (250 bp upstream to 500 bp downstream of the transcription start site) defined by the Roadmap Epigenomics Project. The interactions are considered significant if the binomial test indicated a false discovery rate  $< 1 \times 10^{-567}$  and modified to account for the differences in the cell lines used in this study.

### Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

### Data availability

The Harvard-Emory ECG Database (HEEDB) dataset used in this study is available under restricted access to protect sensitive health data. Access can be obtained by becoming a credentialed user, completing the required "CITI Data or Specimens Only Research" training, and signing the data use agreement via the host platform at <https://bdsf.io/content/heedb/5.0/>. The MIMIC-IV-ECG dataset is available at <https://physionet.org/content/mimic-iv-ecg/1.0/>. The PTB-XL dataset is available at <https://physionet.org/content/ptb-xl/1.0.3/>. The CODE-

15 dataset is available at <https://zenodo.org/records/4916206>. The Chapman dataset is available at <https://physionet.org/content/ecg-arrhythmia/1.0.0/>. The CPSC-2018 dataset is available at <https://physionet.org/content/challenge-2020/1.0.2/sources/>. The ECG and genotype data of the UK Biobank are available under restricted access and require application approval from the UK Biobank (<https://www.ukbiobank.ac.uk/enable-your-research/apply-for-access>). Full GWAS summary statistics used in this study are available at the FinnGen database (version: R8, <https://www.finnngen.fi>) and the GWAS catalog (<https://www.ebi.ac.uk/gwas/>). Source data are provided with this paper.

## Code availability

Our framework was implemented using Python 3.9.20 and the Pytorch 2.5.1. The operating system version is Ubuntu 22.04.2. More details can be found in the “Methods” section, Supplementary Information, and the code repository (<https://github.com/biomed-AI/ECG-LFM>). The Mendelian randomization (MR) analyses were performed by the R package TwoSampleMR (version 0.5.5). The gene association analysis was performed by using MAGMA (version 1.10) (third-party code at <https://ctg.cncr.nl/software/magma>) and the FUMA platform (<https://fuma.ctglab.nl/>). The Linkage disequilibrium score regression (LDSC) analysis was performed by third-party code at <https://github.com/bulik/ldsc>. The colocalization analysis was performed using the R package COLOC (version 5).

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## Author contributions

H.Z., Y.Y., and S.L. conceived the basic idea. S.L. designed the algorithm and built the model. Q.W. and Y.C. collected the data, pre-processed the data, and carried out benchmark experiments. S.L. and Z.L. participated in the case analysis. S.L., H.Z., and Y.Y. prepared the figures and wrote the manuscript. All authors performed literature searches, verified the underlying data, edited the text, critically read the manuscript, and approved the final version for submission. H.Z. provided final edits to the manuscript.

## Competing interests

The authors declare no competing interests.

## Additional information

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